

02443 Stochastic Simulation

Day 4: Variance reduction methods

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Agenda

- 1 Note on the Pareto distribution
- 2 Variance reduction methods
- 3 Exercise 5

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Motivation: When simulation becomes difficult

In many simulation studies, a small number of rare observations may contribute disproportionately to the quantity of interest. Even if extremely large observations occur rarely, they may contribute heavily to moments such as the mean and variance.

Consider a Pareto distribution with density

$$f(x) = \frac{k}{\beta} \left(\frac{x}{\beta} \right)^{-k-1}, \quad x \geq \beta.$$

Contribution to the j th moment

We can analyze moment distributions through the function

$$G_j(x) = \frac{\int_{\beta}^x t^j f(t) dt}{\int_{\beta}^{\infty} t^j f(t) dt} = \frac{\int_{\beta}^x t^j f(t) dt}{\mathbb{E}[X^j]}.$$

The quantity $G_j(x)$ measures the fraction of the j th moment contributed by outcomes less than or equal to x .

For heavy-tailed distributions, convergence of $G_j(x)$ can be surprisingly slow.

The Pareto distribution: A cautionary example

We calculate the first moment distribution of a Pareto-distributed random variable:

$$\int_{\beta}^x tf(t)dt = \frac{\beta k}{k-1} \left[1 - \left(\frac{x}{\beta} \right)^{-k+1} \right].$$

Since

$$E(X) = \frac{\beta k}{k-1},$$

it follows that

$$G_1(x) = 1 - \left(\frac{x}{\beta} \right)^{-k+1}.$$

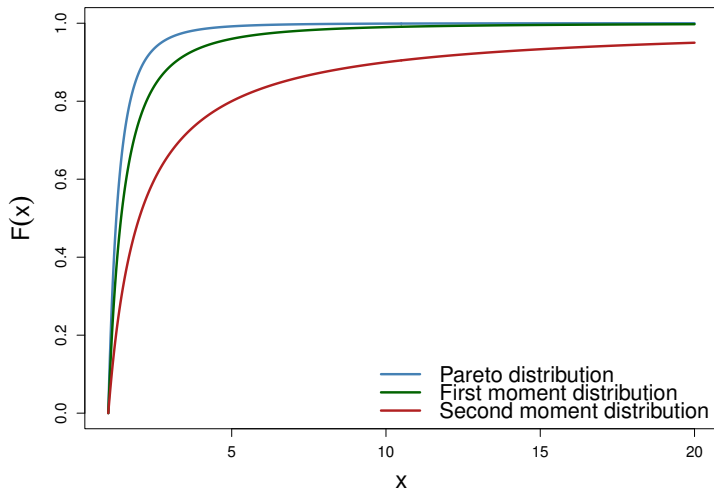
Notice that $G_1(x)$ is itself the distribution function of a Pareto distribution with shape parameter $k-1$.

Important consequence The distribution governing the contribution to the mean has a heavier tail than the original distribution. As a result, extremely large observations can dominate the estimation of moments and substantially increase simulation uncertainty.

The Pareto distribution: Moment distributions (1)

We can plot and tabulate the moment distributions of the Pareto distribution:

Pareto Distribution and Moment Distributions



The Pareto distribution: Moment distributions (2)

For a Pareto distribution with $\beta = 1$ and $k = 2.05$, recall that

$$F(t) = 1 - t^{-k}, \quad G_1(t) = 1 - t^{-k+1}, \quad G_2(t) = 1 - t^{-k+2}.$$

t	$F(t)$	$G_1(t)$	$G_2(t)$
2	0.7585	0.5170	0.0341
10	0.9911	0.9109	0.1190
100	0.9999	0.9921	0.2057
844.5	$1 - 10^{-6}$	0.9992	0.2860

Observation

Even if we simulate 10^6 observations, we should not expect to obtain a reliable estimate of the variance.

Although almost all observations are smaller than 844.5:

$$P(X \leq 844.5) \approx 1 - 10^{-6},$$

these observations contribute only about 29% of the second moment.

Lessons learned

The Pareto example illustrates an important limitation of simulation.

Simulation is not a substitute for understanding.

Heavy tails and rare events

- Rare observations may contribute disproportionately to moments such as the mean and variance.
- Large sample sizes do not automatically guarantee accurate estimates.
- Confidence intervals can be misleading if important regions of the distribution are sampled only rarely.

Lessons learned

Practical implications

- Always combine simulation with theoretical analysis whenever possible.
- Use knowledge of the model to identify important or unusual events.
- Be particularly cautious when studying rare-event probabilities or heavy-tailed distributions.
- Consider specialized simulation techniques when crude Monte Carlo becomes inefficient.

This motivates the study of *variance reduction methods*, whose purpose is to obtain more accurate estimates using the same computational effort.

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Overview

Main idea

Exploit additional information about the problem, such as:

- Analytical properties of the model.
- Correlation between random variables.
- Structural knowledge of the sampling space.

Methods considered in this lecture

- Antithetic variables
- Control variates
- Stratified sampling
- Importance sampling
- Common random numbers

The basics

Suppose we wish to estimate a quantity $\theta = \mathbb{E}[X]$. We can then generate n sample $\{X_i\}_{i=1}^n$ and apply the crude Monte Carlo estimator:

$$\hat{\theta}_{\text{MC}} = \bar{X}.$$

The precision of the estimator is determined by its variance:

$$\mathbb{V}[\hat{\theta}_{\text{MC}}] = \mathbb{V}[\bar{X}] = \frac{\mathbb{V}[X_i]}{n}.$$

A straightforward way to improve precision is therefore to increase the number of simulated observations n .

Alternative approach

Rather than increasing the computational effort, we can often construct a new estimator with a smaller variance.

Example: Estimating an integral

Consider the integral

$$\theta = \int_0^1 e^x dx.$$

This can be interpreted as an expectation by introducing $U \sim U(0, 1)$:

$$\theta = \int_0^1 e^x \cdot 1 dx = \mathbb{E}[e^U].$$

Monte Carlo estimator

Generate independent observations $U_1, U_2, \dots, U_n \sim U(0, 1)$ and define $X_i = e^{U_i}$. The crude Monte Carlo estimator is then

$$\hat{\theta}_{\text{MC}} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \sum_{i=1}^n e^{U_i}.$$

This estimator uses no variance reduction techniques and therefore serves as a natural benchmark.

Analytical benchmark

Before studying variance reduction, let us examine the exact solution.

The integral can be evaluated analytically:

$$\theta = \int_0^1 e^x dx = e - 1 \approx 1.72.$$

For the crude Monte Carlo estimator, the estimator uses the random variables $X_i = e^{U_i}$ with expectation

$$\mathbb{E}[X_i] = \mathbb{E}[e^{U_i}] = e - 1.$$

and variance

$$\mathbb{V}[X_i] = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2 = \frac{e^2 - 1}{2} - (e - 1)^2 \approx 0.2420.$$

The variance follows from:

$$\mathbb{E}[X_i^2] = \mathbb{E}[e^{2U_1}] = \int_0^1 e^{2x} dx = \frac{e^2 - 1}{2}$$

and will serve as our reference point when evaluating variance reduction methods.

Antithetic variables: The idea

Main idea

Use antithetic variables to generate dependent (negatively correlated) random variables and formulate a new estimator with reduced variance.

Suppose that X_i is a monotone function of the uniform random variable U_i . For the integral example, we have $f(x) = \exp(x)$ such that

$$X_i = f(U_i) = e^{U_i}.$$

Instead of using only U_i , we also use its antithetic counterpart $1 - U_i$ to generate a new random variable:

$$Y_i = \frac{f(U_i) + f(1 - U_i)}{2}.$$

The random variable has the same mean value as X_i ,

$$\mathbb{E}[Y_i] = \frac{1}{2} (\mathbb{E}[f(U_i)] + \mathbb{E}[f(1 - U_i)]) = \frac{1}{2} (\theta + \theta) = \theta.$$

Antithetic variables: The estimator

We then introduce the new Monte Carlo estimator:

$$\hat{\theta}_{AV} = \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n \frac{f(U_i) + f(1 - U_i)}{2} = \frac{1}{n} \sum_{i=1}^n \frac{e^{U_i} + e^{1-U_i}}{2}.$$

This estimator is also unbiased

$$\mathbb{E}[\hat{\theta}_{AV}] = \mathbb{E}[\bar{Y}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] = \theta.$$

Observation

The computational effort required to compute $\hat{\theta}_{AV}$ is similar to that required for $\hat{\theta}_{MC}$. We generate the same number uniform numbers (and thus observations) as before while potentially obtaining a substantially smaller variance of the new estimator.

Antithetic variables: Analytical results

For this simple example, the effect of antithetic variables can be analysed analytically.

To determine the estimator variance, we calculate the variance of Y_i

$$\begin{aligned}\mathbb{V}[Y_i] &= \mathbb{V}\left[\frac{f(U_i) + f(1 - U_i)}{2}\right] = \mathbb{V}\left[\frac{e^{U_i} + e^{1-U_i}}{2}\right] \\ &= \frac{1}{4}\mathbb{V}[e^{U_i}] + \frac{1}{4}\mathbb{V}[e^{1-U_i}] + \frac{1}{2}\text{Cov}(e^{U_i}, e^{1-U_i}).\end{aligned}$$

The covariance between $f(U_i)$ and $f(1 - U_i)$ is derived as

$$\begin{aligned}\text{Cov}(f(U_i), f(1 - U_i)) &= \text{Cov}(e^{U_i}, e^{1-U_i}) \\ &= \mathbb{E}[e^{U_1}]\mathbb{E}[e^{1-U_1}] - \mathbb{E}[e^{U_1}]^2 = e - (e - 1)^2 = 3e - e^2 - 1 \approx -0.2342.\end{aligned}$$

Therefore,

$$\mathbb{V}[Y_i] = \frac{1}{2}(0.2420 - 0.2342) = 0.0039.$$

The covariance between $f(U_i)$ and $f(1 - U_i)$ is strongly negative, resulting in a dramatic reduction in the variance of Y_i compared to the variance of X_i .

Antithetic variables: Performance comparison

Crude Monte Carlo

The standard Monte Carlo estimator provides an unbiased estimate of the integral, but the variance may be relatively large. The estimator is based on the variables X_i with variance 0.2420.

Antithetic variables

The antithetic estimator remains unbiased while introducing negative correlation between paired observations. As a result, the estimator is based on the variables Y_i with variance 0.0039.

Key observation

For this example, the variance is reduced by approximately 98%.

This improvement is achieved with essentially the same computational effort as the crude Monte Carlo estimator.

Interpretation

Reducing the variance by a factor of 50 is equivalent to increasing the simulation budget by a factor of 50 when using the crude Monte Carlo method. In other words, antithetic variables can provide the same precision as a much larger simulation, almost for free.

Antithetic variables in more complex models

Assume that the observations X_i depend on several variables:

$$X_i = f(\mathbf{U}_i).$$

If f is monotone in each of its coordinates, then you use the antithetic variables to generate $f(\mathbf{1} - \mathbf{U}_i)$ and base a Monte Carlo estimator on the random variables

$$Y_i = \frac{f(\mathbf{U}_i) + f(\mathbf{1} - \mathbf{U}_i)}{2}.$$

In the queueing system exercise

How can we apply the method of antithetic variables?

Control variates: The idea

Main idea

Generate dependent random variables with known mean and formulate a new estimator with reduced variance.

Assume that we have some random variables Z_i with known mean: $\mathbb{E}[Z_i] = \mu_Z$. We then construct new random variables

$$Y_i = X_i + c(Z_i - \mu_Z).$$

Note that

$$\mathbb{E}[Y_i] = \mathbb{E}[X_i] = \theta.$$

Variance

The variance of Y_i is

$$\mathbb{V}[Y_i] = \mathbb{V}[X_i] + c^2\mathbb{V}[Z_i] + 2c\text{Cov}(X_i, Z_i).$$

Control variates: The idea

Choosing

$$c = -\frac{\text{Cov}(X_i, Z_i)}{\mathbb{V}[Z_i]}$$

minimizes the variance and yields

$$\mathbb{V}[Y_i] = \mathbb{V}[X_i] - \frac{\text{Cov}(X_i, Z_i)^2}{\mathbb{V}[Z_i]}.$$

The stronger the correlation between X_i and Z_i , the larger the variance reduction.

The estimator

We formulate a Monte Carlo estimator based on the observations Y_i :

$$\hat{\theta}_{\text{CV}} = \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n (X_i + c(Z_i - \mu_Z)).$$

This estimator is also unbiased

$$\mathbb{E}[\hat{\theta}_{\text{CV}}] = \mathbb{E}[\bar{Y}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] = \theta.$$

Control variates: Example

Apply U_i as a control variate such that we base our estimator on the observations:

$$Y_i = X_i + c \left(U_i - \frac{1}{2} \right),$$

where we recall that $X_i = e^{U_i}$. We know that $\mathbb{E}[U_i] = 1/2$.

To determine the optimal value of c , we need

$$\text{Cov}(X_i, U_i) = \text{Cov}(e^{U_i}, U_i) \approx 0.14086.$$

The optimal coefficient is therefore

$$c = -\frac{\text{Cov}(e^{U_i}, U_i)}{\mathbb{V}[U_i]} \approx -\frac{0.14086}{1/12} = -1.69.$$

Using this value of c reduces the variance dramatically and produces an estimator with essentially the same variance as the antithetic estimator.

Control variates: Example

Important remark

In practice, we often do not know the covariance between the random variables X_i and the control variates, but rather we need to estimate it empirically.

With the optimal coefficient $c = -1.69$, we base our estimator on observations with variance

$$\mathbb{V}[Y_i] = 0.0039.$$

Stratified sampling

Main idea

Rather than sampling from the entire sample space uniformly, divide the sample space into smaller regions (strata) and sample from each region separately.

For the integral example, we may divide the interval $[0, 1]$ into ten equally sized strata:

$$[0, 0.1], [0.1, 0.2], \dots, [0.9, 1]$$

and generate one observation from each stratum:

$$U_{i,j} \sim U\left(\frac{j-1}{10}, \frac{j}{10}\right), \quad j = 1, \dots, 10.$$

Using the observations

$$Y_i = \frac{1}{10} \sum_{j=1}^{10} e^{U_{i,j}}.$$

we formulate the estimator

$$\hat{\theta}_{\text{SS}} = \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Stratified sampling

Advantages

- Guarantees coverage of the entire sample space.
- Reduces sampling variability.
- Often easy to implement.

A natural question is: how many strata should be used?

Importance sampling

Main idea

Suppose we wish to estimate

$$\theta = E(h(X)) = \int h(x)f(x)dx.$$

Rather than sampling from the density f , we sample from another density g . Provided that $g(x) > 0$ whenever $f(x) > 0$, we may write

$$\theta = \int h(x)\frac{f(x)}{g(x)}g(x)dx = \mathbb{E}\left[h(Y)\frac{f(Y)}{g(Y)}\right],$$

where Y has density g .

This leads to the estimator

$$\hat{\theta}_{\text{IS}} = \frac{1}{n} \sum_{i=1}^n h(Y_i) \frac{f(Y_i)}{g(Y_i)}.$$

Importance sampling

Advantages

- Rare but influential events are sampled more often.
- The estimator remains unbiased through the weight f/g .
- A suitable choice of g can dramatically reduce variance.

Importance sampling is particularly useful for rare-event simulation.

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Exercise 5: Variance reduction methods

Part 1

Estimate the integral

$$\int_0^1 e^x dx$$

by simulation using the crude Monte Carlo estimator.

- Use, for example, an estimator based on 100 samples.
- Present the result as both a point estimate and a confidence interval.

Part 2

Estimate the integral

$$\int_0^1 e^x dx$$

using antithetic variables and comparable computational resources.

Exercise 5 (continued)

Part 3

Estimate the integral

$$\int_0^1 e^x dx$$

using a control variate and comparable computational resources.

Part 4

Estimate the integral

$$\int_0^1 e^x dx$$

using stratified sampling and comparable computational resources.

Part 5

Use control variates to reduce the variance of the estimator in Exercise 4 (Poisson arrivals).

Exercise 5 (continued)

Part 6

Demonstrate the effect of using common random numbers in Exercise 4 when comparing:

- Poisson arrivals,
- A renewal process with hyperexponential inter-arrival times.

Remark: You may need to rethink and modify parts of your simulation code.

Exercise 5 (continued)

Part 7

Let $Z \sim N(0, 1)$. Estimate

$$P(Z > a)$$

using the crude Monte Carlo estimator.

Then apply importance sampling using a normal distribution with mean a and variance σ^2 .

- Start with $\sigma^2 = 1$.
- Experiment with different values of a (e.g., $a = 2$ and $a = 4$).
- Experiment with different sample sizes.
- If time permits, investigate other values of σ^2 .

Finally, discuss the efficiency of the methods.

Exercise 5 (continued)

Part 8

Use importance sampling with

$$g(x) = \lambda e^{-\lambda x}$$

to estimate

$$\int_0^1 e^x dx.$$

Try to identify the optimal value of λ by analysing the variance of

$$\frac{h(X)f(X)}{g(X)}$$

and verify your conclusions through simulation.

Exercise 5 (continued)

Part 9

For the Pareto distribution, derive the importance sampling estimator for the mean using the first moment distribution as the sampling distribution.

- Is this approach meaningful?
- Can it be generalized?
- Based on this insight, can you improve the choice of $g(x)$ in Part 8?

You do not need to implement the method, provided that you can explain what should happen theoretically.